Boyi Li

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Education

Zhejiang University Sep. 2023 – May. 2027
Bachelor of Science in Computer Engineering, Dual Degree with UIUC *Haining, Zhejiang*

- **Relevant Coursework:** Data Structures, Analog Signal Processing, Computer Systems & Programming, Linear Algebra with Computational Applications, Basic Discrete Mathematics, Probability with Engineering Application, etc.
- **GPA:** 4.09/4.3

University of Illinois at Urbana-Champaign (UIUC) Sep. 2023 – May. 2027
Bachelor of Science in Computer Engineering, Dual Degree with ZJU *Champaign, Illinois*

- **Relevant Coursework:** Machine Learning, Applied Machine Learning, Artificial Intelligence, Computer Systems Engineering, Digital Systems Laboratory, Intro Differential Equations, etc.
- **GPA:** 3.86/4

Publications

*Equal contribution.

Yifan Shen, Jian Xu, **Boyi Li**, Yuner Zhang, Tianjiao Yu, Bingxuan Li, Houze Yang, Rushi Wang, Xu Cao. ChronoVision: Temporal Reasoning via Latent State Reconstruction. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2026.

Yifan Shen*, **Boyi Li***, Meihuan Huang*, Yuanzhe Liu*, Xu Cao*, Jinyang Jin, Zhengyuan Li, Anglin Liu, Junho Kim, Jingyuan Zhu, Lan Fangzhou, Jianguo Cao, Jintai Chen, Ismini Lourentzou, James M. Rehg. Decoding Children's Gait Behavior. In *European Conference on Computer Vision (ECCV)*. 2026.

Boyi Li, Yifan Shen, Houze Yang, Xu Cao, Guojun Yun, Li Gao, Turong Chen, Long Xu, Jianguo Cao, Meihuan Huang. The 1st AI Children Challenge . In *Computer Vision and Pattern Recognition (CVPR) Workshop on Computer Vision for Children (CV4CHL)*. 2026.

Boyi Li*, Zhonghan Zhao*, Der-Horng Lee, Gaoang Wang. Adaptive Graph Pruning for Multi-Agent Communication. In *European Conference on Artificial Intelligence (ECAI)*. 2025. **Spotlight Paper**.

Zhonghan Zhao*, Wenhao Chai*, Xuan Wang*, **Boyi Li**, Shengyu Hao, Shidong Cao, Tian Ye, Gaoang Wang. See and Think: Embodied Agents in Virtual Environments. In *European Conference on Computer Vision (ECCV)*. 2024.

Research Experience

Rehg Lab, UIUC **Advisor: Prof. James M. Rehg** Sep. 2025 – Present
Research Intern *Champaign, Illinois*

Conducted research on Multimodal LLMs.

- Developed Cognitive Supersensing, a cognition-inspired training framework that equips MLLMs with human-like visual imagination for abstract reasoning. Trained CogSense-8B under the proposed framework.
- CogSense achieves up to 33.5% improvement over strong baselines on cognitive tasks and maintains competitive general abilities at the same time.
- Proposed ChronoVision, a comprehensive framework for visual temporal reasoning that guide MLLMs focus toward fine-grained dynamic key regions and understand the sequential relationship between frames.
- ChronoVision overpasses mainstream open/closed-sourse MLLMs on defined image ordering tasks, achieving 22.6% improvement compared to the strongest baseline and maintaining competitive general abilities at the same time.

Conducted research on Computer Vision for Healthcare.

- Explored the domain of fine-grained pediatric gait analysis from video for clinically relevant motion understanding.
- Curated the largest pediatric gait dataset. Benchmarked mainstream MLLMs and gait analysis models on the dataset.
- Proposed ChildrenGait-Video, a state-of-the-art pediatric-centric gait analysis models for Edinburgh Visual Gait Score (EVGS) scoring.

CVNext Lab, Zhejiang University **Advisor: Prof. Gaoang Wang** Sep. 2023 – Aug 2025
Research Intern *Haining, Zhejiang*

Conducted research on Embodied AI.

- Proposed STEVE, a comprehensive and visionary embodied agent in the Minecraft virtual environment, with a corresponding dataset that includes vision-environment, knowledge question-answering, and skill-code pairs.
- STEVE achieves up to 1.5x faster unlocking key tech trees and 2.5x quicker in block search tasks.

Conducted research on Multi-Agent Systems.

- Identified inefficiencies in communication-heavy multi-agent systems and proposed Adaptive Graph Pruning (AGP), a framework that jointly optimizes agent quantity and communication topology.
- AGP achieves an increase in performance of 2.58% - 9.84% across six mainstream benchmarks with a decrease in token consumption of 90%+.

ICI Lab, Zhejiang University **Advisor: Prof. Liuqing Chen** Jun. 2024 – Jul. 2024
Research Intern *Hangzhou, Zhejiang*

Conducted research on Human Computer Interaction.

- Investigated sketch-based interaction methods for lowering the barrier to 3D design in virtual reality environments.
- Proposed a system combining freehand VR input with model recognition to enable automatic reconstruction of structured 3D geometry.
- Conducted preliminary user evaluations showing improved efficiency and usability for novice users.

Service

Workshop Co-Organizer CVPR 2026
The 2nd CVPR 2026 Workshop on Computer Vision for Children

- Co-organized the 2nd CVPR Workshop on Computer Vision for Children (CV4CHL), helping shape workshop themes, challenge design, and community engagement.
- Led the organization of the 1st AI for Children Challenge, including dataset curation and evaluation design.

Projects

Unix-like Operating System

- **Role:** Project lead. Primarily responsible for the UART, VirtIO, and File System components.
- Developed a Unix-like operating system supporting user program execution, virtual memory management, and system calls, using C, RISC-V assembly, and Sv39 paging.
- Implemented virtual memory subsystem with multi-level page tables, demand paging, and secure memory isolation for user processes.
- Implemented process abstraction and ELF loader, enabling dynamic program loading, trap-based user/kernel transitions.

BioElectro Glove: Mental Health Screening through Connective Tissue Signals

- **Role:** Tech Lead. Responsible for product architecture design and model training.
- Based on fascia theory, this project explores depression diagnosis via hand bioelectrical signals, leveraging the physiological link between emotional states, connective tissue health, and oxidative stress, enabling a non-invasive and repeatable detection method.
- The project won a Gold Award in the China International College Students' Innovation Competition.

Honors and Awards

Dean's List University of Illinois Urbana-Champaign, Spring 2026

Gold Award (Top 0.01%) China International College Students' Innovation Competition (CICSIC), 2025

Gold Award (Top 0.01%) Singapore Division Contest of China International College Students' Innovation Competition (CICSIC-SG), 2025

Bronze Award (Top 0.01%) China International College Students' Innovation Competition (CICSIC), 2024

Honorable Mention Mathematical Contest In Modeling (MCM), 2024

Third-Class Academic Excellence Scholarship Zhejiang University, 2024 – 2025

Student Outstanding Academic Achievement Award Zhejiang University, 2024 – 2025

Student Innovation and Entrepreneurship Award Zhejiang University, 2023 – 2024, 2024 – 2025

Languages and Skills

Chinese: Native

English: Proficient (TOEFL 5/6 (100/120))

Programming Languages: C/C++, Python, C#, JavaScript

Hardware & Embedded Systems: FPGA, Intel Quartus